

Morgan Patel

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EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Cut the signal's daily turnover from 34% to 21% with a cost-aware position smoother, keeping 90% of gross returns and lowering estimated slippage by a third.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so no look-ahead information leaked into training.
- Tested the signal on six years of tick-level data using purged walk-forward splits with an embargo, so that no future information leaked into training.
- Presented the signal, its capacity estimate and failure cases to the portfolio managers, who approved a small live allocation for the next quarter.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2020 - Aug 2021

- Responsible for the lab's simulation pipeline and for helping other students run their experiments on the shared cluster.
- I worked on some statistics for a paper and I also helped my advisor with the math, which I think made the paper better.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation while also maintaining the lab website, organizing the weekly reading group, grading for two courses, reviewing papers for my advisor, helping new students set up the cluster, attending several conferences, and writing documentation for other lab tools, which together increased downloads, improved lab productivity, and made the group's research more visible to the broader statistics community.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Leveraged cutting-edge quantitative synergies and next-generation modeling paradigms to unlock alpha-driven innovation.
- Reduced forecast error from 0.20 to 0.25, a 40% improvement, with a new asymmetric loss function for volatility spikes.
- Ported the VX-9 stack onto QRT under the Helix program, coordinating with the Atlas and Borealis teams.

Kaggle Market Prediction Competition | Team of 3 | Python | Mar 2023 - Jun 2023

- Worked with various people on multiple aspects of the competition to help the team make progress.
- Was involved in the process of helping to support the validation work that was being done by the team during the competition.
- Wrote the team's feature-selection script using permutation importance, trimming 900 candidate features to 300 without losing validation score.

SKILLS

Programming: Python, R, SQL, C++

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch

Infrastructure: Kubernetes, Haskell, Kafka